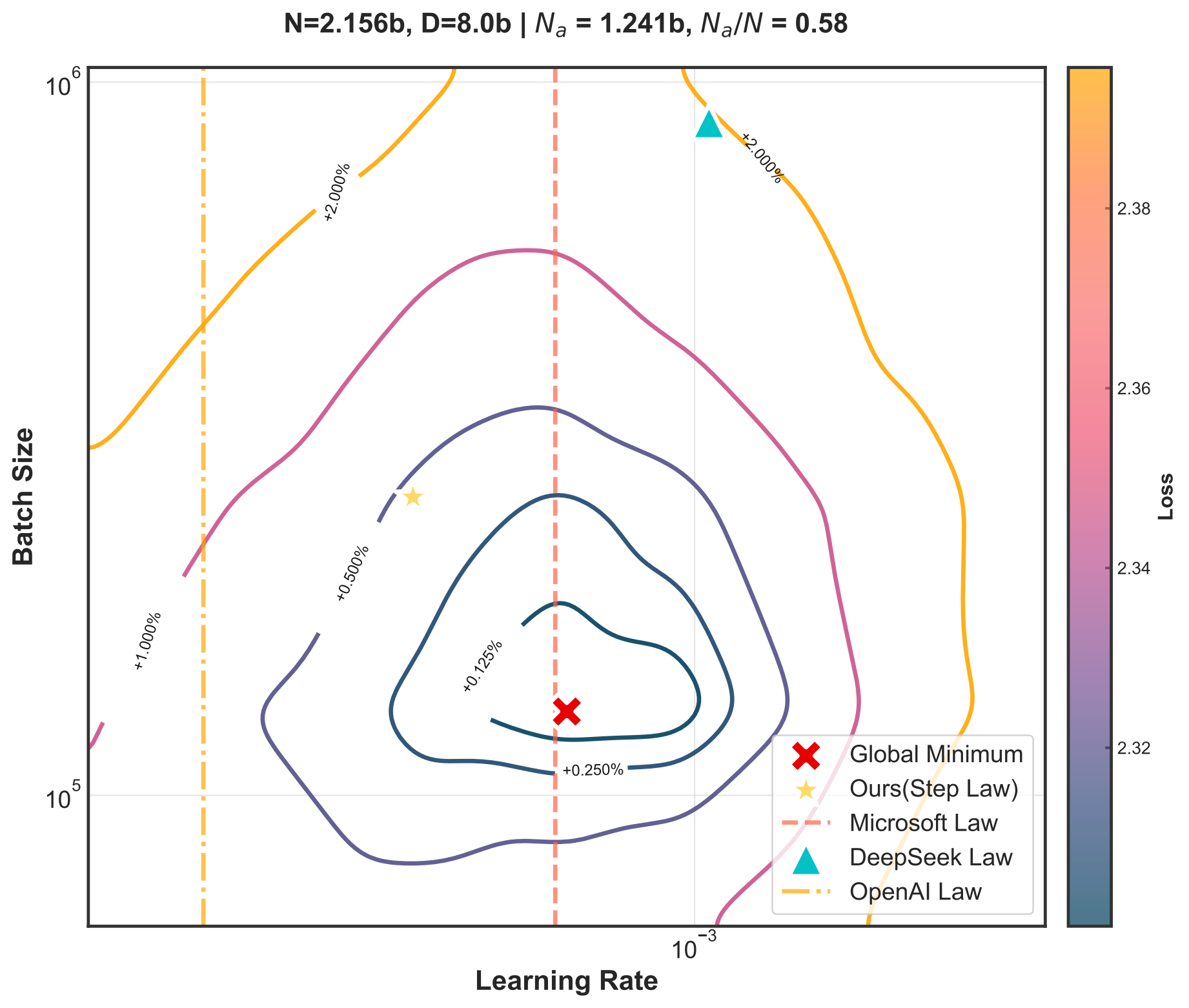
N=2.156b, D=8.0b | N_a = 1.241b, N_a/N = 0.58
Batch Size
Learning Rate
Loss
10^6
10^5
10^{-3}
2.38
2.36
2.34
2.32
+2.000%
+2.000%
+1.000%
+0.500%
+0.125%
+0.250%
Global Minimum
Ours(Step Law)
Microsoft Law
DeepSeek Law
OpenAI Law